

AI Assisted Coding

ASSIGNMENT 4.1

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PROBLEM STATEMENT 1 : Customer Email Classification

A company receives a large number of customer emails every day and wants to automatically classify them into the following categories:

- Billing
- Technical Support
- Feedback
- Others

Instead of training a new machine learning model, the company decides to use prompt engineering techniques with an existing large language model.

Tasks

- 1. Prepare five short sample emails, each belonging to one of the above categories.**

Email 1 – Billing

Hello team, I was charged twice for my subscription this month. Please help me get a refund.

Email 2 – Technical Support

My app crashes every time I try to log in after the recent update. Kindly assist.

Email 3 – Feedback

I really like the new interface. It's much easier to navigate now. Great job!

Email 4 – Others

I would like to know if your services are available in international locations.

Email 5 – Billing

My invoice shows an incorrect amount compared to what I selected during checkout

2. Write a zero-shot prompt to classify a given email into one of the categories without providing any examples.

Zero-Shot Prompting

Prompt

Classify the following customer email into one of these categories:

Billing, Technical Support, Feedback, Others.

Email:

"I was charged twice for my subscription this month. Please help me."

Output

Billing

Observation

- The model correctly identifies the category without any examples.
- Zero-shot prompting works well for clear and simple emails.
- Accuracy may reduce if the email is vague or contains mixed issues.

3. Write a one-shot prompt by including one labeled email example and ask the model to classify a new email.

Zero-Shot Prompting

Prompt

Classify the following customer email into one of these categories:

Billing, Technical Support, Feedback, Others.

Email:

"I was charged twice for my subscription this month. Please help me."

Output

Billing

Observation

- The model correctly identifies the category without any examples.
- Zero-shot prompting works well for clear and simple emails.
- Accuracy may reduce if the email is vague or contains mixed issues.

4. Write a few-shot prompt by including two or three labeled email examples and ask the model to classify a new email.

Few-Shot Prompting

Prompt

Classify customer emails into the categories:

Billing, Technical Support, Feedback, Others.

Examples:

Email: "I was billed extra for last month's usage."

Category: Billing

Email: "The website is not loading on my laptop."

Category: Technical Support

Email: "I appreciate the fast response from your support team."

Category: Feedback

Now classify this email:

Email: "Do you offer student discounts for your services?"

Output

Others

Observation

- Few-shot prompting gives the best accuracy.
- The model clearly understands each category from multiple examples.

- Most effective for ambiguous or new types of customer emails.

5. Compare the outputs obtained using zero-shot, one-shot, and few-shot prompting techniques and briefly comment on their effectiveness

Prompting Type	Accuracy	Reliability	Observation
Zero-Shot	Medium	Moderate	Works for simple, direct emails
One-Shot	High	Better	Example improves clarity
Few-Shot	Very High	Best	Most accurate and consistent

Overall Observation

- Few-shot prompting is the most effective for customer email classification.
- As the number of examples increases, the model's understanding improves.
- Prompt engineering can successfully replace training a new ML model for such tasks.

PROBLEM STATEMENT 2: Intent Classification for Chatbot Queries

A company wants to deploy a chatbot to handle customer queries. Each query must be classified into one of the following intents: Account Issue, Order Status, Product Inquiry, or General Question using prompt engineering techniques.

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